TEXORA · LECTURE SHEET

Machinery Audit

Daily process control assumes the machines are fit; the machinery audit verifies that assumption systematically. This sheet covers the audit method — checklists, scoring, department comparison — and how audit findings convert into maintenance priorities before they become quality losses.

YM-4101 · PROCESS CONTROL IN SPINNING · TOPIC 12 · SPECIALIZATION — YARN MANUFACTURING

1. Why audit machines separately from product

Product tests are lagging indicators: by the time sliver U% or ends down deteriorate, the worn roller or opened setting has been damaging output for days. A **machinery audit** is a leading indicator — a structured inspection of machine condition against a checklist of mechanical and technological points, scored so that departments and dates can be compared. It answers a different question from process control: not "is the product right?" but "is the machine still capable of making it right all shift, all week?"

2. The audit method

- **Checklist per machine type:** fixed points covering settings (gauged, not estimated), condition of working elements (clothing, cots, aprons, travellers, bearings), drives and speeds (tachometer-verified against the data sheet), drafting system, suction and cleaning systems, safety guards, and lubrication state.
- **Scoring:** each point scores full marks, part marks, or zero; critical points (e.g. cylinder–flat setting out of tolerance) may be weighted or made pass/fail gates regardless of total score.
- **Sampling:** all machines of a type over an audit cycle, or a rotating sample deep-checked; the audit team includes a technologist and a maintenance fitter so findings are diagnosed, not just noted.
- **Follow-through:** every deduction carries an action, an owner and a date; the next audit verifies closure. An audit without follow-through is tourism.

WORKED EXAMPLE — AN AUDIT SCORED

A spinning audit uses 500 points distributed by department importance. Scores achieved:

DEPARTMENT	SCORED	POSSIBLE	%
Blowroom	66	80	82.5
Carding	86	100	86.0
Comber & preparation	47	60	78.3
Drawframe	54	60	90.0
Simplex	42	50	84.0
Ring frame	88	100	88.0
Winding	48	50	96.0
Mill total	**431**	**500**	**86.2%**

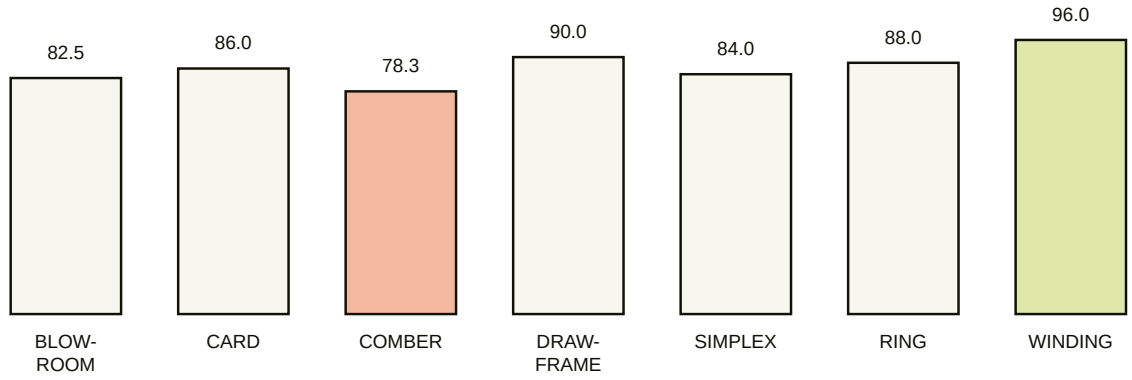


Figure 1 — Department audit scores of the worked example (bar heights to scale). The mill average of 86.2% conceals the comber department at 78.3% — audits are read department first, average second.

Reading the result: winding (96%) and drawframe (90%) need only routine care. The comber department at 78.3% is the priority — in the detailed findings, two combers carried detaching-roller damage and one had a failed top-comb set, exactly the faults Topic 6 teaches process control to detect from noil and U% data. The audit and the process data corroborate each other; where they disagree, believe both and look again.

3. What the checklist examines, by area

AREA	REPRESENTATIVE AUDIT POINTS
Settings	Roller settings gauged vs data sheet; cylinder–flat; comber detaching setting; clearer settings vs order specification

Working elements	Clothing condition and grinding state; cots (cracks, hardness, eccentricity); aprons; travellers within life
Drives and speeds	Spindle/doffer/rotor speeds tachometer-checked; belts and tapes condition; gear backlash
Material path	Guides aligned, creels free, suction effective, ducts clean, no sharp edges or rust on contact surfaces
Housekeeping & safety	Fly accumulation, oil leaks, guards fitted, fire points accessible
Records	Setting sheets current, maintenance history complete, previous audit actions closed

4. Audit frequency and grades

A common pattern: full department audits quarterly, with monthly spot audits on the lowest-scoring department until it recovers. Mills often grade: ≥ 90% excellent, 80–90% acceptable with actions, below 80% — intervention (maintenance shutdown planning, intensified process testing on that department's output). The grade belongs to the department at a date; tracking the same department across audits shows whether maintenance is winning or losing.

Glossary

Machinery audit — a scored, checklist inspection of machine condition and settings. **Deduction** — points lost at a checklist item, each carrying an action. **Leading indicator** — a measure that deteriorates before the product does. **Setting sheet (data sheet)** — the standard settings document against which the audit gauges. **Spot audit** — a short, focused re-audit of one department or machine group. **Closure** — verification at the next audit that a previous action was completed and effective.

Key points

• The audit measures machine capability; process control measures product. Each catches what the other misses.
• Worked example: mill 431/500 = 86.2%, but the comber department's 78.3% is the actionable figure.
• Checklists cover settings, working elements, drives/speeds, material path, housekeeping and records.
• Every deduction needs an action, owner and date — and closure verified at the next audit.
• Persistent low scorers get spot audits and intensified product testing until they recover.

Practice

1. An audit scores departments: A 92/120, B 145/160, C 68/80. Calculate each % and the overall %, and state which department gets the first intervention.
2. A carding department scores 88%, yet sliver neps are rising. Give two explanations that reconcile the audit and the product data.
3. Why does the audit re-verify the closure of previous actions instead of simply re-scoring the machine?

Answer key

1. A = 92/120 = **76.7%**; B = 145/160 = **90.6%**; C = 68/80 = **85.0%**. Overall = (92+145+68)/(120+160+80) = 305/360 = **84.7%**. First intervention: department A — lowest score, below the 80% intervention grade.
2. (1) The audit is a snapshot; clothing or settings may have deteriorated since, or the fault is intermittent (e.g. one card of many). (2) The checklist may be missing the causal point — nep rise can come from the material (immature mixing) while machines are genuinely sound; the audit measures machines, not cotton. Both possibilities direct different actions, so product localisation (which cards, since when) is checked first.
3. Because unscored promises repeat: without closure verification, the same deductions reappear audit after audit while reports look complete. Closure checking converts the audit from a scoring exercise into a maintenance management system.

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